

30-Week AI/ML Engineer Roadmap (4 Hours Daily)

DAY 1 — BUILDING AI/ML ENGINEER MINDSET

Before You Start... Read This Carefully.

If you are reading this page, one thing is clear:

- 👉 You don't want to blindly follow courses anymore.
- 👉 You want **clarity & real direction**.
- 👉 You want to become an **AI/ML Engineer with understanding**, not confusion.

And let me tell you something honestly...

Most students fail not because AI is hard.
They fail because they start the **wrong way**.

The Biggest Mistake Students Make

They:

- ❌ Start coding directly
- ❌ Buy random courses
- ❌ Copy YouTube tutorials
- ❌ Memorize algorithms
- ❌ Do projects without understanding

After few months... They say:
"AI is not for me."

But the truth is:

- 👉 The problem is NOT AI. Its in **Approach**

Hello Boss... Let's Talk Reality.

You cannot build a building without a foundation.
Then why are you trying to build AI without Learning Real Math?

Real AI Engineers don't start with coding.
They start with Understanding **Math**.

Without this:
Coding becomes confusion.

Our Route is Different (And Proven)

Most roadmaps say:

“Start Python → ML → DL”

But we don't follow that blindly.

This KIT is designed based on:

- ✓ Real AI Engineers' experiences
- ✓ Real industry requirements for JOB
- ✓ Real student mistakes
- ✓ Real execution methods

You Are Not Alone

At the end of this KIT,

you will get access to a **Telegram Community** where:

- ✓ AI Engineers share referrals
- ✓ Students clear doubts
- ✓ Members help each other
- ✓ You can build real connections
- ✓ You can also give referrals once you grow

You don't just learn here.

You become part of a growing AI community.

You will learn —

Theoretically + Practically

with Videos , Best Websites, Mini Projects & Major Projects

- ◆ Linear Algebra
- ◆ Calculus
- ◆ Probability Theory
- ◆ Statistics
- ◆ Optimization
- ◆ Information Theory
- ◆ Set Theory & Logic
- ◆ Graph Theory
- ◆ High-Level Topics for Advanced AI Research

- ✓ Machine Learning (ML)
- ✓ Deep Learning (DL)
- ✓ Build Real Projects (ML + DL)

- ✓ Specializations (NLP, CV, GenAI)
- ✓ MLOps (Production & Scaling)
- ✓ Competitive Exposure & Portfolio Building
- ✓ Job Preparation (40+ LPA Roles)

- 📦 Tools & Libraries You Must Master
- 📌 What to Learn After Math
- 🧱 Data Engineering Foundations
- 🔒 AI Ethics, Fairness & Privacy
- 🧠 Paper Reading & AI Research Skills
- 🧮 Advanced Mathematics (Optional)
- 📦 Prompt Engineering & LLM Ops
- 🧱 Systems Thinking & AI Infrastructure
- 👤 Soft Skills + Domain Knowledge

- ✓ Final Checklist
- 🚀 Execution Phase

All structured in a **clear step-by-step order**
so you always know what to learn next
without confusion.

🎯 Final Goal

Not just learning AI...
👉 Becoming a real AI Engineer step-by-step.

Week 1 — Linear Algebra (AI Engineer Foundation Week)

🎯 Weekly Goal

Understand all Linear Algebra topics and subtopics with visual clarity.
Move to the next topic only after complete understanding.

📌 Topics & Subtopics to Complete

1. Scalars, Vectors, Matrices, Tensors

- Notation
- Dimensions

- Representation
- Broadcasting concept

2. Vector Operations

- Addition
- Subtraction
- Scalar multiplication
- Dot product
- Cross product

3. Matrix Operations

- Matrix addition
- Matrix multiplication
- Transpose
- Identity matrix

4. Matrix Inverses & Determinants

- Inverse of matrix
- Singular vs non-singular matrices
- Determinant properties

5. Linear Independence

- Rank
- Null space
- Column space

6. Systems of Linear Equations

- Gaussian elimination
- Consistency of equations

7. Span & Basis

- Basis vectors
- Vector spaces

8. Orthogonality

- Orthogonal vectors
- Vector projection
- Gram-Schmidt process

9. Eigenvalues & Eigenvectors

- Eigendecomposition
- Diagonalization

10. Matrix Factorization

- LU Decomposition
- QR Decomposition

11. Singular Value Decomposition (SVD)

- Concept
- Low-rank approximation

12. Principal Component Analysis (PCA)

- Covariance matrix
 - Eigenvectors as principal axes
-

Important Instruction

Not everyone understands from the same source.

You may follow your own learning resources.

But after completing each topic and subtopic, move ahead **only when the concept is fully understood visually and conceptually.**

Suggested Learning Resources

YouTube Playlists

- 3Blue1Brown – Essence of Linear Algebra

<https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr>

- Khan Academy – Linear Algebra

<https://www.khanacademy.org/math/linear-algebra>

- MIT OpenCourseWare – Linear Algebra

<https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/>

Helpful Websites

- Khan Academy Practice

<https://www.khanacademy.org/math/linear-algebra>

- MIT Practice Problems

<https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/pages/assignments/>

Practice Tasks

- Identify scalars, vectors, matrices from examples
- Perform vector addition and dot product manually
- Multiply matrices step-by-step

- Solve systems of equations
 - Find determinant and inverse of small matrices
 - Observe matrix transformations visually
 - Write short notes for each topic
-

Weekly Self-Check Questions

- What is the difference between a vector and a matrix?
 - Why is dot product important in AI?
 - When is a matrix invertible?
 - What does rank tell us?
 - Why are eigenvalues important in PCA?
 - How does Linear Algebra help AI systems process data?
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Where Linear Algebra is Used in AI

- Neural Networks
 - Word Embeddings
 - PCA (Dimensionality Reduction)
 - Attention Mechanisms
 - Computer Vision
 - Data Representation in ML Models
-

Mini Project

Create a visual explanation of how PCA works:

- Show how data points are transformed
- Explain eigenvectors visually
- Write your understanding in simple terms

Goal: Build conceptual clarity.

End of Week Outcome

You should be able to:

- Understand vectors & matrices confidently
- Solve basic linear algebra problems
- Explain eigenvalues conceptually

- Understand how PCA reduces dimensions
 - Build strong foundation for Machine Learning
-

Week 2 — Calculus (Learning Mechanism of AI Models)

Weekly Goal

Understand how calculus helps AI models learn, optimize, and minimize errors using derivatives and gradients.

Topics & Subtopics to Complete

1. Limits & Continuity

- Basic limit theory
- Function continuity

2. Differentiation

- Power rule
- Product rule
- Quotient rule
- Chain rule

3. Partial Derivatives

- Multivariable functions
- $\partial f / \partial x$, $\partial f / \partial y$

4. Gradient Vectors

- Gradient meaning
- Direction of steepest ascent

5. Jacobian & Hessian Matrices

- First-order derivatives
- Second-order derivatives
- Curvature understanding

6. Directional Derivatives

- Optimization along directions

7. Taylor Series Expansions

- Function approximation
- Role in optimization

8. Backpropagation Algorithm

- Chain rule application
- Error minimization logic

9. Integration (Optional)

- Area under curves
 - Expectation basics
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Suggested Learning Resources

YouTube Playlists

- Khan Academy – Calculus

<https://www.khanacademy.org/math/calculus-1>

- 3Blue1Brown – Calculus Series

<https://www.youtube.com/playlist?list=PLZHQObOWTQDNPOjrT6KVlfJuKtYTftqH6>

- MIT OpenCourseWare – Calculus

<https://ocw.mit.edu/courses/18-01-single-variable-calculus-fall-2006/>



Helpful Websites

- Paul's Online Math Notes – Calculus

<https://tutorial.math.lamar.edu>

- Khan Academy Practice

<https://www.khanacademy.org/math/calculus-1>

- MIT Practice Problems

<https://ocw.mit.edu>



Practice Tasks

- Solve limits manually
- Differentiate basic functions
- Apply chain rule problems
- Compute partial derivatives
- Visualize gradient direction
- Understand Taylor approximations
- Write summary notes

Weekly Self-Check Questions

- What is a limit and why is continuity important?
 - Why are derivatives used in AI optimization?
 - What does the gradient represent?
 - Difference between Jacobian and Hessian?
 - How does backpropagation use calculus?
-

Where Calculus is Used in AI

- Backpropagation
 - Gradient Descent
 - Loss Minimization
 - Model Optimization
 - Neural Network Training
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Mini Project

Explain visually how gradient descent works:

- Show how error reduces step-by-step
- Plot a simple curve
- Describe how derivatives guide optimization

Goal: Build intuition on how AI models learn.

End of Week Outcome

You should be able to:

- Understand derivatives confidently
- Interpret gradients conceptually
- Relate calculus to AI learning
- Understand how optimization works in models

Week 3 — Probability Theory (AI's Decision-Making Foundation)

Weekly Goal

Understand how probability helps AI models handle uncertainty, make predictions, and reason with data.

Topics & Subtopics to Complete

1. Sample Space & Events

- Set theory basics
- Outcomes
- Venn diagrams

2. Probability Axioms

- Addition rule
- Multiplication rule

3. Conditional Probability

- Understanding $P(A|B)$
- Real-life examples

4. Bayes' Theorem

- Prior
- Likelihood
- Posterior

5. Independence

- Independent events
- Dependent events

6. Random Variables

- Discrete vs Continuous
- Mapping outcomes to numbers

7. Probability Distributions

Discrete Distributions

- Bernoulli
- Binomial
- Poisson

Continuous Distributions

- Uniform
- Normal (Gaussian)
- Exponential

8. Expectation & Variance

- Mean ($E[X]$)

- Variance
- Standard deviation

9. Covariance & Correlation

- Relationship between variables

10. Law of Large Numbers

- Empirical vs theoretical probability

11. Central Limit Theorem

- Normal distribution behavior
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Suggested Learning Resources

YouTube Playlists

- Khan Academy – Probability & Statistics
<https://www.khanacademy.org/math/statistics-probability>
 - StatQuest – Probability Basics
<https://www.youtube.com/c/joshstarmer>
 - MIT OpenCourseWare – Probability
<https://ocw.mit.edu/courses/18-05-introduction-to-probability-and-statistics-spring-2014/>
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Helpful Websites

- Khan Academy Practice
<https://www.khanacademy.org/math/statistics-probability>
 - ProbabilityCourse.com
<https://www.probabilitycourse.com>
 - MIT OCW Assignments
<https://ocw.mit.edu>
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Practice Tasks

- Identify sample spaces for events
- Solve conditional probability problems
- Apply Bayes' theorem
- Plot probability distributions

- Compute mean and variance
 - Interpret correlation between variables
-

Weekly Self-Check Questions

- What is conditional probability?
 - How does Bayes' theorem update beliefs?
 - Difference between discrete & continuous distributions?
 - Why is normal distribution important in AI?
 - How does probability help AI decision-making?
-

Where Probability is Used in AI

- Naive Bayes Algorithm
 - Generative Models
 - Probabilistic Modeling
 - LLM Predictions
 - AI Reasoning Systems
-

Mini Project

Analyze a real-life scenario using probability:

- Define events
- Apply Bayes' theorem
- Explain prediction results

Goal: Build intuition on how AI makes decisions under uncertainty.

End of Week Outcome

You should be able to:

- Understand probability concepts clearly
- Interpret distributions confidently
- Apply Bayes' theorem
- Relate probability to AI reasoning and predictions

Week 4 — Statistics (Data Analysis, Evaluation & Generalization)

Weekly Goal

Understand how statistics helps analyze data, evaluate AI models, and make reliable decisions from data.

Topics & Subtopics to Complete

1. Descriptive Statistics

- Mean
- Median
- Mode
- Variance
- Skewness
- Kurtosis

2. Inferential Statistics

- Estimation
- Confidence intervals

3. Sampling Techniques

- Random sampling
- Stratified sampling

4. Hypothesis Testing

- Null hypothesis vs Alternate hypothesis
- p-value
- z-test
- t-test

5. Correlation vs Causation

- Pearson correlation
- Spearman correlation

6. Maximum Likelihood Estimation (MLE)

- Parameter estimation basics

7. Bayesian Inference

- Prior
- Posterior
- Marginalization

8. Bias–Variance Tradeoff

- Underfitting
- Overfitting
- Generalization

9. Confidence Intervals

- Interpretation of parameter bounds

10. Outlier Detection

- IQR method
 - Z-score method
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Suggested Learning Resources

YouTube Playlists

- StatQuest – Statistics Fundamentals
<https://www.youtube.com/c/joshstarmar>
 - Khan Academy – Statistics
<https://www.khanacademy.org/math/statistics-probability>
 - MIT OpenCourseWare – Statistics
<https://ocw.mit.edu>
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Helpful Websites

- Khan Academy Practice
<https://www.khanacademy.org/math/statistics-probability>
 - Statistics How To
<https://www.statisticshowto.com>
 - MIT OCW Practice Problems
<https://ocw.mit.edu>
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Practice Tasks

- Calculate mean, median, variance
- Interpret confidence intervals
- Perform hypothesis testing examples
- Identify correlation in datasets

- Detect outliers using IQR method
 - Write summary notes
-

Weekly Self-Check Questions

- Difference between descriptive & inferential statistics?
 - What does p-value indicate?
 - Why is sampling important?
 - What is bias–variance tradeoff?
 - How do outliers affect models?
-

Where Statistics is Used in AI

- Model Evaluation
 - Performance Diagnostics
 - Hypothesis Validation
 - Uncertainty Quantification
 - Data Analysis for ML Models
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Mini Project

Analyze a small dataset manually:

- Calculate descriptive statistics
- Identify outliers
- Interpret correlations
- Summarize insights

Goal: Understand how data evaluation supports AI decision-making.

End of Week Outcome

You should be able to:

- Interpret statistical measures
- Understand hypothesis testing
- Evaluate data distributions
- Relate statistics to AI model performance

Week 5 — Optimization (How AI Models Learn)

Weekly Goal

Understand how AI models reduce errors and improve performance using optimization techniques.

Topics & Subtopics to Complete

1. Objective / Loss Functions

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Cross-Entropy Loss
- Hinge Loss

2. Gradient Descent

- Batch Gradient Descent
- Stochastic Gradient Descent (SGD)
- Mini-batch Gradient Descent
- Momentum

3. Learning Rate

- Learning rate concept
- Scheduling
- Annealing

4. Convex Functions

- Global minima vs Local minima
- Convex optimization basics

5. Constrained Optimization

- Lagrange multipliers

6. Stochastic Optimization

- SGD
- Adam
- RMSprop
- Adagrad

7. Second-order Optimization

- Newton's Method
- Hessian Matrix
- BFGS

8. Regularization Techniques

- L1 Regularization (Lasso)

- L2 Regularization (Ridge)
 - Dropout
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Suggested Learning Resources

YouTube Playlists

- StatQuest – Gradient Descent & Optimization
<https://www.youtube.com/c/joshstarmer>
 - Andrew Ng – Optimization Concepts
<https://www.coursera.org/learn/machine-learning>
 - MIT OpenCourseWare – Optimization
<https://ocw.mit.edu>
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Helpful Websites

- Distill.pub – Optimization Visual Guides
<https://distill.pub>
 - Google ML Crash Course
<https://developers.google.com/machine-learning/crash-course>
 - MIT OCW Practice Materials
<https://ocw.mit.edu>
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Practice Tasks

- Compare different loss functions
 - Visualize gradient descent steps
 - Understand effect of learning rate
 - Identify convex vs non-convex functions
 - Study optimizer differences
 - Write summary notes
-

Weekly Self-Check Questions

- What is the role of a loss function?
- Why is gradient descent important?

- How does learning rate affect convergence?
 - Difference between SGD and Adam?
 - Why is regularization needed?
-



Where Optimization is Used in AI

- Model Training
 - Neural Network Convergence
 - Hyperparameter Tuning
 - Loss Minimization
 - Performance Improvement
-



Mini Project

Create a visual explanation of gradient descent:

- Show how error decreases step-by-step
- Compare learning rates
- Explain optimizer behavior

Goal: Build intuition on how AI models learn through optimization.



End of Week Outcome

You should be able to:

- Understand optimization fundamentals
- Interpret loss functions
- Explain gradient descent clearly
- Relate optimization to model performance

Week 6 — Information Theory (Used in NLP, LLMs, Autoencoders)



Weekly Goal

Understand how information is measured, how uncertainty is reduced, and how AI models evaluate predictions using information theory.

Topics & Subtopics to Complete

1. Entropy

- Information content
- Uncertainty measurement

2. Cross Entropy

- Loss function for classification
- Relation to entropy

3. Kullback-Leibler (KL) Divergence

- Difference between probability distributions

4. Mutual Information

- Shared information between variables

5. Bits, Encoding & Compression

- Shannon entropy
- Huffman coding basics

6. Perplexity

- Evaluation metric for language models
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Suggested Learning Resources

YouTube Playlists

- StatQuest – Information Theory
<https://www.youtube.com/c/joshstarmer>
 - 3Blue1Brown – Information Theory Concepts
<https://www.youtube.com>
 - MIT OpenCourseWare – Information Theory
<https://ocw.mit.edu>
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Helpful Websites

- Distill.pub – Visual Explanations
<https://distill.pub>
- Khan Academy – Probability & Information
<https://www.khanacademy.org>

- MIT OCW Notes
<https://ocw.mit.edu>
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Practice Tasks

- Calculate entropy for simple examples
 - Compare two distributions using KL divergence
 - Understand cross-entropy loss visually
 - Interpret mutual information examples
 - Write summary notes
-

Weekly Self-Check Questions

- What does entropy measure?
 - Why is cross-entropy used in classification?
 - What is KL divergence used for?
 - How does perplexity evaluate language models?
 - Why is information theory important in NLP?
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Where Information Theory is Used in AI

- NLP Models (Transformers)
 - GAN Training
 - LLM Evaluation
 - BERT & GPT Models
 - Autoencoders
-

Mini Project

Explain visually how entropy changes with probability:

- Compare high vs low uncertainty cases
- Show cross-entropy effect on predictions
- Write your interpretation

Goal: Build intuition on uncertainty and prediction evaluation.

End of Week Outcome

You should be able to:

- Understand entropy & uncertainty
- Interpret cross-entropy loss
- Compare probability distributions
- Relate information theory to NLP & LLMs

Week 7 — Set Theory & Logic (Foundations of Algorithms & AI Reasoning)

Weekly Goal

Understand how sets, functions, and logic form the foundation for AI reasoning, algorithms, and decision-making systems.

Topics & Subtopics to Complete

1. Set Operations

- Union
- Intersection
- Difference
- Complement

2. Functions & Relations

- One-to-one functions
- Onto functions
- Mappings between sets

3. Logic

- Propositional logic
- Truth tables
- Logical statements

4. Boolean Algebra

- Logic gates
- Simplification rules

5. Quantifiers

- $\forall$ (For all)
 - $\exists$ (There exists)
-

Suggested Learning Resources

YouTube Playlists

- Neso Academy – Discrete Mathematics
<https://www.youtube.com/c/NesoAcademy>
 - Khan Academy – Logic & Set Theory
<https://www.khanacademy.org>
 - MIT OpenCourseWare – Discrete Mathematics
<https://ocw.mit.edu>
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Helpful Websites

- Brilliant.org – Logic & Sets Practice
<https://brilliant.org>
 - Paul's Online Math Notes
<https://tutorial.math.lamar.edu>
 - MIT OCW Lecture Notes
<https://ocw.mit.edu>
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Practice Tasks

- Solve set operation problems
 - Draw Venn diagrams
 - Create truth tables
 - Simplify Boolean expressions
 - Identify logical statements
-

Weekly Self-Check Questions

- Difference between union and intersection?
 - What is a one-to-one function?
 - Why are truth tables important?
 - How does Boolean algebra help algorithms?
 - What do quantifiers represent?
-



Where Set Theory & Logic is Used in AI

- Symbolic AI
 - Search Algorithms
 - Logical Reasoning Systems
 - Rule-Based Systems
 - Knowledge Representation
-



Mini Project

Create a logical decision system:

- Define rules using Boolean logic
- Draw truth tables
- Explain how AI can make decisions using logic

Goal: Build understanding of reasoning in AI systems.



End of Week Outcome

You should be able to:

- Perform set operations confidently
- Interpret logical expressions
- Build truth tables
- Understand reasoning foundations in AI

Week 8 — Graph Theory (Used in Knowledge Graphs, Transformers)



Weekly Goal

Understand how graph structures represent relationships and how AI systems use graphs for connections, reasoning, and learning patterns.



Topics & Subtopics to Complete

1. Graphs

- Nodes

- Edges
- Graph types (Directed, Undirected)
- Adjacency Matrix

2. Trees & DAGs

- Tree structure basics
- Directed Acyclic Graphs (DAGs)
- Decision Trees (Concept Level)

3. Graph Traversals

- Breadth-First Search (BFS)
- Depth-First Search (DFS)

4. Shortest Path Algorithms

- Dijkstra Algorithm
- A* Algorithm

5. Connectivity

- Connected Components
- Spanning Trees



Suggested Learning Resources

YouTube

- freeCodeCamp — Graph Data Structures & Algorithms
<https://www.youtube.com/c/Freecodecamp>
- William Fiset — Graph Theory (Visual Explanations)
<https://www.youtube.com/c/WilliamFiset-videos>



Helpful Websites

- Visualgo (Graph Visual Learning)
<https://visualgo.net>
 - GeeksforGeeks — Graph Theory Concepts
<https://www.geeksforgeeks.org/graph-data-structure-and-algorithms>
 - MIT OpenCourseWare
<https://ocw.mit.edu>
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Practice Tasks

- Draw graphs with nodes and edges
 - Represent graphs using adjacency matrix
 - Perform BFS traversal manually
 - Perform DFS traversal manually
 - Solve shortest path example step-by-step
 - Identify connected components in graphs
-

Weekly Self-Check Questions

- What is the difference between graph and tree?
 - When do we use BFS vs DFS?
 - Why are shortest path algorithms important?
 - What is a DAG and why is it useful?
 - How do graphs represent relationships in AI?
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Where Graph Theory is Used in AI

- Knowledge Graphs (Google Search, Recommendation Systems)
 - Transformers (Attention Relationship Mapping)
 - Graph Neural Networks (GNNs)
 - Social Network Analysis
 - Path Optimization Systems
-

Mini Project

Create a simple relationship graph:

- Create nodes (people, products, or places)
- Create connections between them
- Explain how AI can use this graph for recommendations or reasoning

Goal: Understand relationship-based data representation.

End of Week Outcome

You should be able to:

- Understand graph structures clearly

- Perform BFS & DFS conceptually
- Understand shortest path logic
- Relate graph theory concepts to AI systems

Bonus Week — Advanced Math Topics for AI Research (Optional)

Weekly Goal

Get high-level awareness of advanced mathematical concepts used in cutting-edge AI research.

(No deep mastery required. Focus on concept-level understanding.)

Topics to Explore

1. Manifold Learning

- Nonlinear dimensionality reduction
- t-SNE (concept)
- UMAP (concept)

Used In:

Representation Learning • Visualization • Advanced ML Research

2. Riemannian Geometry

- Geometry on curved spaces
- Optimization on curved surfaces (concept level)

Used In:

Advanced GAN Optimization • LLM Training Research • Deep Learning Theory

3. Functional Analysis

- Function spaces concept
- Convergence of functions (basic idea)

Used In:

LLM Architecture Theory • Advanced Neural Network Research

4. Measure Theory

- Probability in continuous spaces
- Infinite-dimensional probability (concept level)

Used In:

Advanced ML Theory • Probabilistic Modeling Research

Suggested Learning Resources

YouTube (Concept Understanding)

- StatQuest — Advanced Math for ML (Concept Videos)

<https://www.youtube.com/c/joshstarmer>

- 3Blue1Brown — Advanced Math Intuition

<https://www.youtube.com/c/3blue1brown>

- Yannic Kilcher — AI Research Concepts

<https://www.youtube.com/c/YannicKilcher>

Helpful Websites

- Papers With Code (Research Understanding)

<https://paperswithcode.com>

- Distill.pub (Visual Research Explanations)

<https://distill.pub>

- arXiv (Research Papers — Optional Reading)

<https://arxiv.org>

Practice Tasks

- Watch concept videos for each topic
 - Note where each topic is used in AI
 - Try to relate concepts to ML / DL systems
 - Write short summaries
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Self-Check Questions

- Why do we need nonlinear dimensionality reduction?
 - Why is geometry important in deep learning research?
 - Why does advanced AI need strong math theory?
 - Where are these topics used in modern AI research?
-



Where These Are Used in AI

- Advanced LLM Research
 - Generative AI Research
 - Deep Learning Theory
 - Cutting-edge ML Research Labs
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Mini Project

Pick one advanced topic and create a simple explanation:

- What is it?
- Where is it used?
- Why is it important in future AI?

Goal: Build research awareness mindset.



End of Bonus Week Outcome

You should be able to:

- Recognize advanced AI math terms
- Understand where advanced math is used in AI
- Build curiosity towards research-level AI topics

Week 9 —Stage 1 — Machine Learning Foundations



Weekly Goal

Understand core Machine Learning concepts and how data is prepared, processed, and evaluated before building ML models.



Topics & Subtopics to Complete

1. What is Machine Learning

- Definition of ML
 - Types of ML:
 - Supervised Learning
 - Unsupervised Learning
 - Semi-Supervised Learning
 - Reinforcement Learning
-

2. Data Preprocessing

- Handling Missing Data
 - Data Scaling
 - Encoding (Label Encoding, One-Hot Encoding)
 - Data Normalization
-

3. Feature Engineering

- Feature Selection
 - Feature Extraction
 - Polynomial Features (Concept Level)
-

4. Evaluation Metrics

- Accuracy
 - Precision
 - Recall
 - F1 Score
 - ROC Curve
 - AUC Score
 - Confusion Matrix
-



Suggested Learning Resources

YouTube

- StatQuest — Machine Learning Basics
<https://www.youtube.com/c/joshstarmer>
 - Google ML Crash Course
<https://developers.google.com/machine-learning/crash-course>
 - Andrew Ng — ML Concepts
<https://www.coursera.org/learn/machine-learning>
-

Helpful Websites

- Google ML Crash Course
<https://developers.google.com/machine-learning/crash-course>
 - Kaggle Learn (ML Basics)
<https://www.kaggle.com/learn>
 - Scikit-Learn Documentation (Concept Reference)
<https://scikit-learn.org>
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Practice Tasks

- Identify ML type for real-world problems
 - Practice data cleaning examples
 - Understand scaling and normalization differences
 - Compare evaluation metrics using examples
 - Interpret confusion matrix
-

Weekly Self-Check Questions

- Difference between supervised and unsupervised learning?
 - Why is preprocessing important?
 - When do we use precision vs recall?
 - What does ROC curve show?
 - Why is feature engineering important?
-

Where ML Foundations Are Used

- Data Prediction Systems
 - Recommendation Engines
 - Fraud Detection
 - Customer Analytics
 - Pattern Recognition Systems
-

Mini Project

Take a sample dataset (CSV or sample data):

- Identify features and target
- Identify ML problem type
- Decide preprocessing steps
- Decide evaluation metrics

Goal: Understand how ML pipeline starts before model building.

End of Week Outcome

You should be able to:

- Understand ML types clearly
- Understand data preprocessing importance
- Interpret evaluation metrics
- Understand ML workflow from data to model evaluation

Week 10 — Supervised Learning Algorithms

Weekly Goal

Understand how supervised learning algorithms learn from labeled data and how they are used for prediction and classification problems.

Algorithms & Concepts to Complete

1. Linear Regression

- Cost Function (MSE)
 - Gradient Descent (Concept)
 - Overfitting (Basic Idea)
-

2. Logistic Regression

- Sigmoid Function
 - Probability Output
 - Cross-Entropy Loss
-

3. K-Nearest Neighbors (KNN)

- Euclidean Distance

- Choosing Optimal K
 - Distance-Based Prediction
-

4. Decision Trees

- Gini Impurity
 - Entropy
 - Information Gain
 - Overfitting in Trees
-

5. Random Forest

- Ensemble Learning
 - Bagging Concept
 - Multiple Trees Voting
-

6. Support Vector Machines (SVM)

- Hyperplanes
 - Margin Maximization
 - Kernel Trick (Linear, RBF — Concept Level)
-

Suggested Learning Resources

YouTube

- StatQuest — ML Algorithms Explained
<https://www.youtube.com/c/joshstارmer>
 - Andrew Ng — Machine Learning Course
<https://www.coursera.org/learn/machine-learning>
 - Krish Naik — ML Practical Concepts
<https://www.youtube.com/c/KrishNaik>
-

Helpful Websites

- Scikit-Learn Documentation
<https://scikit-learn.org>
- Kaggle Learn — Machine Learning
<https://www.kaggle.com/learn>

- Google ML Crash Course

<https://developers.google.com/machine-learning/crash-course>



Practice Tasks

- Identify regression vs classification problems
 - Understand prediction logic of each algorithm
 - Compare algorithm use cases
 - Understand bias vs variance examples
 - Interpret model predictions
-



Weekly Self-Check Questions

- Difference between Linear & Logistic Regression?
 - Why is K important in KNN?
 - Why do Decision Trees overfit?
 - How does Random Forest reduce overfitting?
 - Why do SVM use kernels?
-



Where Supervised Learning is Used

- Sales Prediction
 - Spam Detection
 - Fraud Detection
 - Medical Diagnosis
 - Customer Churn Prediction
-



Mini Project

Choose a real-world problem and:

- Identify if it is regression or classification
- Choose suitable algorithm
- Explain why you selected it

Goal: Build algorithm selection thinking.

End of Week Outcome

You should be able to:

- Understand supervised ML algorithms clearly
- Know when to use each algorithm
- Understand prediction logic conceptually
- Prepare for ML model implementation in next weeks

Week 11 — Unsupervised Learning Algorithms

Weekly Goal

Understand how unsupervised learning finds hidden patterns, groups, and structures in data without labeled outputs.

Algorithms & Concepts to Complete

1. K-Means Clustering

- Centroids
 - Cluster Formation
 - Inertia
 - Elbow Method
-

2. Hierarchical Clustering

- Agglomerative Clustering (Concept)
 - Dendrograms
 - Linkage Criteria
-

3. DBSCAN

- Density-Based Clustering
 - Epsilon (ϵ)
 - Minimum Points (MinPts)
 - Noise / Outlier Detection
-

4. PCA (Dimensionality Reduction)

- Covariance Matrix
 - Eigen Decomposition (Concept)
 - Explained Variance
-

Suggested Learning Resources

YouTube

- StatQuest — Clustering & PCA
<https://www.youtube.com/c/joshstarmer>
 - Krish Naik — Unsupervised Learning
<https://www.youtube.com/c/KrishNaik>
 - 3Blue1Brown — PCA Intuition
<https://www.youtube.com/c/3blue1brown>
-

Helpful Websites

- Scikit-Learn Clustering Documentation
<https://scikit-learn.org>
 - Kaggle Learn — Feature Engineering & ML
<https://www.kaggle.com/learn>
 - Google ML Crash Course
<https://developers.google.com/machine-learning/crash-course>
-

Practice Tasks

- Identify clustering problems in real life
 - Understand difference between K-Means & DBSCAN
 - Interpret dendrogram structure
 - Understand how PCA reduces dimensions
 - Compare clustering outputs conceptually
-

Weekly Self-Check Questions

- Difference between supervised and unsupervised learning?
 - Why do we use clustering?
 - When is DBSCAN better than K-Means?
 - Why do we use PCA before ML models?
 - What is explained variance?
-



Where Unsupervised Learning is Used

- Customer Segmentation
 - Recommendation Systems
 - Anomaly Detection
 - Pattern Recognition
 - Feature Reduction in ML Pipelines
-



Mini Project

Choose a dataset scenario and:

- Identify possible clusters
- Decide which clustering algorithm fits best
- Explain why PCA might be used before clustering

Goal: Build pattern recognition thinking.



End of Week Outcome

You should be able to:

- Understand clustering algorithms clearly
- Know when to use each algorithm
- Understand dimensionality reduction conceptually
- Prepare for model tuning and evaluation stages

Week 12 — Model Selection & Tuning



Weekly Goal

Understand how to select the best model, avoid overfitting, and improve model performance using tuning techniques.

Topics & Subtopics to Complete

1. Cross Validation

- K-Fold Cross Validation
 - Stratified K-Fold
-

2. Hyperparameter Tuning Methods

- Grid Search
 - Random Search
-

3. Bias–Variance Tradeoff

- Underfitting
 - Overfitting
 - Model Generalization
-

4. Learning Curves

- Training vs Validation Performance
 - Detecting Overfitting / Underfitting
-

Suggested Learning Resources

YouTube

- StatQuest — Bias vs Variance & Cross Validation
<https://www.youtube.com/c/joshstarmer>

- Krish Naik — Hyperparameter Tuning
<https://www.youtube.com/c/KrishNaik>

- Andrew Ng — Model Evaluation Concepts
<https://www.coursera.org/learn/machine-learning>
-

Helpful Websites

- Scikit-Learn Model Selection Guide
<https://scikit-learn.org>
 - Kaggle Learn — Model Validation
<https://www.kaggle.com/learn>
 - Google ML Crash Course
<https://developers.google.com/machine-learning/crash-course>
-

Practice Tasks

- Understand why we split training and validation data
 - Compare Grid Search vs Random Search
 - Interpret bias vs variance examples
 - Understand learning curve graphs
 - Identify overfitting vs underfitting cases
-

Weekly Self-Check Questions

- Why is cross-validation important?
 - When to use Stratified K-Fold?
 - Difference between Grid Search and Random Search?
 - What causes overfitting?
 - How do learning curves help model tuning?
-

Where Model Selection & Tuning is Used

- ML Model Optimization
 - Hyperparameter Tuning
 - Model Performance Improvement
 - Production ML Systems
 - Real-World Model Deployment
-

Mini Project

- Take a sample ML model scenario and:
- Decide validation strategy

- Select tuning method (Grid or Random Search)
- Explain how you will detect overfitting

Goal: Build model optimization thinking.

End of Week Outcome

You should be able to:

- Understand model validation methods
- Understand hyperparameter tuning basics
- Interpret bias vs variance tradeoff
- Understand how to improve ML model performance

Week 13 — Deep Learning Basics (Neural Network Fundamentals)

Weekly Goal

Understand how neural networks work, how they learn patterns, and how predictions improve using forward and backward learning processes.

Topics & Subtopics to Complete

1. Perceptron

- Basic neuron concept
 - Input → Weight → Output idea
-

2. Activation Functions

- ReLU
 - Sigmoid
 - Tanh
 - Why activation is needed
-

3. Neural Network Architecture

- Input Layer
- Hidden Layer

- Output Layer
 - Neuron connections
-

4. Forward Propagation

- Weighted Sum
 - Activation Function Output
 - Signal Flow Through Network
-

5. Backpropagation

- Chain Rule Concept
 - Gradient Calculation
 - Error Reduction Logic
-

6. Loss Functions

- MSE
 - Cross-Entropy
 - Hinge Loss
-

Suggested Learning Resources

YouTube

- 3Blue1Brown — Neural Networks
<https://www.youtube.com/c/3blue1brown>
 - StatQuest — Neural Network Basics
<https://www.youtube.com/c/joshstarmer>
 - Andrew Ng — Deep Learning Concepts
<https://www.coursera.org/specializations/deep-learning>
-

Helpful Websites

- TensorFlow Neural Network Guide
<https://www.tensorflow.org>
- DeepLearning.ai Learning Resources
<https://www.deeplearning.ai>

- Google ML Crash Course (Neural Networks Section)
<https://developers.google.com/machine-learning/crash-course>
-

Practice Tasks

- Draw a simple neural network diagram
 - Identify input, hidden, output layers
 - Understand activation function curves
 - Understand forward vs backward learning concept
 - Compare loss functions
-

Weekly Self-Check Questions

- What is a perceptron?
 - Why do neural networks need activation functions?
 - What happens in forward propagation?
 - Why is backpropagation important?
 - How do loss functions affect learning?
-

Where Neural Networks Are Used

- Image Recognition
 - Speech Recognition
 - NLP Systems
 - Recommendation Systems
 - Prediction Models
-

Mini Project

Draw and explain a simple neural network:

- Define inputs and outputs
- Explain forward propagation flow
- Explain how error is reduced using backpropagation

Goal: Build strong neural network intuition.

End of Week Outcome

You should be able to:

- Understand neural network structure
- Understand how networks learn from data
- Understand role of activation and loss functions
- Prepare for advanced Deep Learning architectures

Week 14 — Deep Learning Training Techniques

Weekly Goal

Understand how neural networks are trained efficiently, how learning is stabilized, and how overfitting is controlled during training.

Topics & Subtopics to Complete

1. Gradient Descent Variants

- Batch Gradient Descent
 - Stochastic Gradient Descent (SGD)
 - Mini-Batch Gradient Descent
-

2. Weight Initialization

- Why initialization matters
 - Xavier Initialization
 - He Initialization
-

3. Training Stabilization Techniques

- Dropout
 - Batch Normalization
-

4. Training Process Terms

- Epochs
 - Iterations
 - Batch Size
-

5. Overfitting & Regularization

- Overfitting vs Underfitting
 - L1 Regularization
 - L2 Regularization
-

Suggested Learning Resources

YouTube

- StatQuest — Gradient Descent Variants & Regularization
<https://www.youtube.com/c/joshstarmer>
 - DeepLearning.ai — Training Deep Networks
<https://www.coursera.org/specializations/deep-learning>
 - Krish Naik — Deep Learning Training Concepts
<https://www.youtube.com/c/KrishNaik>
-

Helpful Websites

- TensorFlow Training Guide
<https://www.tensorflow.org>
 - PyTorch Training Tutorials
<https://pytorch.org>
 - Google ML Crash Course
<https://developers.google.com/machine-learning/crash-course>
-

Practice Tasks

- Compare Batch vs SGD vs Mini-Batch conceptually
 - Understand effect of batch size on training
 - Study dropout effect on neurons
 - Understand why normalization stabilizes learning
 - Compare L1 vs L2 regularization effects
-

Weekly Self-Check Questions

- Why do we use mini-batch instead of batch training?
 - Why is weight initialization important?
 - How does dropout reduce overfitting?
 - What is batch normalization solving?
 - Difference between L1 and L2 regularization?
-



Where Training Techniques Are Used

- Deep Neural Network Training
 - CNN Model Training
 - NLP Model Training
 - Large Scale Deep Learning Systems
 - Production AI Model Optimization
-



Mini Project

Explain training flow of a neural network:

- Show how batches are processed
- Explain epochs and iterations
- Explain how dropout or regularization prevents overfitting

Goal: Build understanding of real training workflow.



End of Week Outcome

You should be able to:

- Understand deep learning training workflow
- Understand how to stabilize training
- Understand how overfitting is controlled
- Prepare for advanced neural network architectures

Week 15 — Advanced Deep Learning Architectures



Weekly Goal

Understand how advanced neural network architectures handle images, sequences, and language using specialized network designs.

Topics & Subtopics to Complete

1. CNN (Convolutional Neural Networks)

- Convolution Operation
 - Filters / Kernels
 - Strides
 - Padding
 - Pooling (Max / Avg)
-

2. RNN (Recurrent Neural Networks)

- Sequence Modeling
 - Time-Step Processing
 - Vanishing Gradient Problem
-

3. LSTM / GRU

- Long-Term Dependency Handling
 - Input Gate
 - Forget Gate
 - Output Gate
 - GRU Simplified Gating
-

4. Transformers

- Self-Attention Mechanism
 - Positional Encoding
 - Encoder–Decoder Architecture
 - Parallel Processing Advantage
-

5. Attention Mechanism

- Query
 - Key
 - Value
 - Dot-Product Attention
-

Suggested Learning Resources

YouTube

- 3Blue1Brown — Neural Networks & Deep Learning

<https://www.youtube.com/c/3blue1brown>

- Yannic Kilcher — Transformers Explained

<https://www.youtube.com/c/YannicKilcher>

- StatQuest — CNN & RNN Basics

<https://www.youtube.com/c/joshstarmarmer>



Helpful Websites

- TensorFlow Model Architecture Guides

<https://www.tensorflow.org>

- PyTorch Tutorials

<https://pytorch.org/tutorials>

- The Illustrated Transformer (Visual Guide)

<https://jalammar.github.io/illustrated-transformer/>



Practice Tasks

- Understand difference between CNN and ANN
 - Understand how RNN processes sequences
 - Study why LSTM solves vanishing gradient
 - Understand attention mechanism flow
 - Compare CNN vs Transformer use cases
-



Weekly Self-Check Questions

- Why CNN is best for images?
 - Why RNN struggles with long sequences?
 - How LSTM solves memory problem?
 - Why Transformers replaced RNNs in NLP?
 - How attention helps models focus on important information?
-



Where Advanced Networks Are Used

- CNN — Image Recognition, Computer Vision
 - RNN / LSTM — Time Series, Speech, NLP
 - Transformers — LLMs, Chatbots, Translation
 - Attention — NLP, Vision Transformers, LLMs
-

Mini Project

Choose one architecture and explain workflow visually:

- Input → Processing → Output
- Explain why that architecture is used for that problem

Goal: Build architecture selection understanding.

End of Week Outcome

You should be able to:

- Understand advanced deep learning architectures
- Know when to use CNN vs RNN vs Transformers
- Understand attention concept clearly
- Prepare for real deep learning project building

Week 16 — Deep Learning Optimization Algorithms

Weekly Goal

Understand how advanced optimizers improve neural network training speed, stability, and convergence.

Topics & Subtopics to Complete

1. SGD (Stochastic Gradient Descent)

- Basic SGD concept
 - Noise in updates
 - Faster but unstable updates (concept level)
-

2. Adam Optimizer

- Adaptive learning rates
 - Momentum + RMSprop combination
 - Most widely used optimizer in deep learning
-

3. RMSprop

- Adaptive learning rate per parameter
 - Stabilizes gradient updates
-

4. Learning Rate Schedulers

- Step Decay
 - Exponential Decay
 - Reduce on Plateau
 - Warm-up Scheduling (Concept Level)
-

5. Gradient Clipping

- Prevent exploding gradients
 - Stabilize training in deep networks
-



Suggested Learning Resources

YouTube

- StatQuest — Optimizers Explained
<https://www.youtube.com/c/joshstarmar>
 - DeepLearning.ai — Optimization Techniques
<https://www.coursera.org/specializations/deep-learning>
 - Krish Naik — Optimizers Practical Concepts
<https://www.youtube.com/c/KrishNaik>
-



Helpful Websites

- TensorFlow Optimizer Guide
<https://www.tensorflow.org>
- PyTorch Optimization Tutorials
<https://pytorch.org/tutorials>

- Google ML Crash Course

<https://developers.google.com/machine-learning/crash-course>

Practice Tasks

- Understand difference between SGD, Adam, RMSprop
 - Study optimizer use cases
 - Understand why learning rate matters
 - Study effect of learning rate scheduling
 - Understand exploding gradient problem conceptually
-

Weekly Self-Check Questions

- Why Adam is commonly used in deep learning?
 - When is SGD preferred over Adam?
 - Why do we need learning rate scheduling?
 - What problem does gradient clipping solve?
 - How optimizers affect model training speed?
-

Where These Optimizers Are Used

- Deep Neural Network Training
 - CNN & Vision Models
 - NLP Models & LLM Training
 - GAN Training
 - Large-Scale AI Systems
-

Mini Project

Compare optimizers conceptually:

- Choose a DL model scenario
- Decide which optimizer is suitable
- Explain why

Goal: Build optimizer selection understanding.

End of Week Outcome

You should be able to:

- Understand deep learning optimizers clearly
- Know when to use SGD vs Adam vs RMSprop
- Understand learning rate tuning importance
- Understand training stability techniques

Week 17 — Real AI Projects (Beginner Level)

Weekly Goal

Apply Machine Learning and Deep Learning fundamentals to build real beginner-level AI projects and understand complete ML workflow.

Projects & Concepts to Complete

1. House Price Prediction

Algorithms Used:

- Linear Regression

Concept Focus:

- Regression Problems
 - Feature Selection
 - Model Evaluation (MSE, R^2 Concept)
-

2. Titanic Survival Prediction

Algorithms Used:

- Logistic Regression
- Decision Tree

Concept Focus:

- Classification Problems
 - Feature Importance
 - Model Accuracy Evaluation
-

3. MNIST Digit Classification

Algorithms Used:

- MLP (Multi-Layer Perceptron)
- CNN (Concept Level)

Concept Focus:

- Image Classification Basics
 - Neural Network Workflow
 - Training vs Testing Concept
-

4. Spam Email Classifier

Algorithms Used:

- TF-IDF
- Logistic Regression

Concept Focus:

- Text Feature Extraction
 - NLP Basics
 - Binary Classification
-



Suggested Learning Resources

YouTube

- Krish Naik — ML Project Tutorials
<https://www.youtube.com/c/KrishNaik>

- freeCodeCamp — ML Projects
<https://www.youtube.com/c/Freecodecamp>

- Sentdex — ML & AI Projects
<https://www.youtube.com/c/sentdex>
-



Helpful Websites

- Kaggle (Datasets + Notebooks)
<https://www.kaggle.com>
 - UCI ML Repository
<https://archive.ics.uci.edu>
 - Google Colab (Free ML Coding Environment)
<https://colab.research.google.com>
-

Practice Tasks

- Understand problem statement of each project
 - Identify input features and output target
 - Understand which algorithm fits each project
 - Understand evaluation metric used
 - Study dataset structure
-

Weekly Self-Check Questions

- When do we use regression vs classification?
 - Why is MNIST used for DL learning?
 - Why TF-IDF is used in spam detection?
 - How do ML projects follow data → model → evaluation flow?
-

Where These Beginner Projects Are Used

- Real Estate Prediction Systems
 - Risk Prediction Models
 - Document Classification Systems
 - Image Recognition Systems
-

Mini Project

Choose one beginner project and:

- Explain ML pipeline flow
- Explain algorithm choice
- Explain evaluation method

Goal: Build real-world ML project thinking.

End of Week Outcome

You should be able to:

- Understand ML project workflow
- Understand dataset → model → evaluation pipeline
- Understand real ML project structure
- Prepare for intermediate AI projects

Week 18 — Real AI Projects (Intermediate Level)

Weekly Goal

Build intermediate-level AI understanding by learning how real-world AI systems handle images, text, recommendations, and conversations.

Projects & Concepts to Complete

1. Face Recognition System

Tech Used:

- CNN
- OpenCV

Concept Focus:

- Image Feature Extraction
 - Face Detection vs Face Recognition
 - CNN Feature Learning
-

2. Chatbot System

Tech Used:

- RNN
- Transformer (Concept Level)

Concept Focus:

- Sequence Processing
 - Text Understanding
 - Context-Based Response
-

3. Movie Recommendation System

Tech Used:

- Collaborative Filtering
- Cosine Similarity

Concept Focus:

- User Similarity
 - Item Similarity
 - Recommendation Logic
-

4. Sentiment Analysis System

Tech Used:

- Word Embeddings
- LSTM

Concept Focus:

- Text Representation
 - Sequence Learning
 - Emotion / Opinion Detection
-

Suggested Learning Resources

YouTube

- Krish Naik — AI Projects
<https://www.youtube.com/c/KrishNaik>
 - Sentdex — AI + Computer Vision + NLP
<https://www.youtube.com/c/sentdex>
 - freeCodeCamp — NLP & CV Projects
<https://www.youtube.com/c/Freecodecamp>
-

Helpful Websites

- Kaggle (Datasets + Code Examples)
<https://www.kaggle.com>
 - Hugging Face (NLP Models & Tutorials)
<https://huggingface.co>
 - OpenCV Documentation
<https://opencv.org>
-

Practice Tasks

- Understand which project belongs to CV / NLP / Recommender domain
- Understand dataset types used
- Understand model selection logic
- Study real-world application use cases
- Compare traditional ML vs Deep Learning approaches

Weekly Self-Check Questions

- Why CNN is best for face recognition?
 - Why Transformers are replacing RNN in chatbots?
 - How collaborative filtering gives recommendations?
 - Why embeddings are needed in NLP?
-

Where These Intermediate Projects Are Used

- Security & Surveillance Systems
 - Virtual Assistants & Chatbots
 - OTT & E-Commerce Recommendation Engines
 - Social Media Sentiment Analysis
-

Mini Project

Choose one intermediate project and:

- Explain system workflow (Input → Model → Output)
- Explain why that model is used
- Explain real-world business usage

Goal: Build real AI solution design thinking.

End of Week Outcome

You should be able to:

- Understand real AI system architectures
- Know which tech stack fits which problem
- Understand AI system pipeline design
- Prepare for advanced AI projects

Week 19 — Real AI Projects (Advanced Level)

Weekly Goal

Understand how advanced AI systems combine Deep Learning models to solve complex real-world problems across vision, language, and time-series domains.

Projects & Concepts to Complete

1. Image Caption Generator

Deep Learning Used:

- CNN
- LSTM

Concept Focus:

- Image Feature Extraction
 - Sequence Generation
 - Vision + NLP Combination
-

2. Object Detection

Deep Learning Used:

- YOLO
- SSD

Concept Focus:

- Object Localization
 - Bounding Box Prediction
 - Real-Time Detection
-

3. Text Summarization

Deep Learning Used:

- Transformer Models
- BART
- T5

Concept Focus:

- Sequence-to-Sequence Learning
 - Context Understanding
 - Text Compression
-

4. Fake News Detection

Deep Learning Used:

- BERT
- DistilBERT

Concept Focus:

- Contextual Text Understanding
 - NLP Classification
 - Transfer Learning
-

5. Time Series Forecasting**Deep Learning Used:**

- LSTM
- ARIMA (Statistical + ML Approach)

Concept Focus:

- Sequential Data Prediction
 - Trend & Seasonality Detection
-

 Suggested Learning Resources**YouTube**

- Krish Naik — Advanced AI Projects
<https://www.youtube.com/c/KrishNaik>
 - Yannic Kilcher — Advanced AI Models
<https://www.youtube.com/c/YannicKilcher>
 - freeCodeCamp — Deep Learning Projects
<https://www.youtube.com/c/Freecodecamp>
-

 Helpful Websites

- Kaggle — Advanced Datasets
<https://www.kaggle.com>
 - Hugging Face — Transformer Models
<https://huggingface.co>
 - Papers With Code
<https://paperswithcode.com>
-

 Practice Tasks

- Understand multi-model pipeline (CNN + LSTM etc.)
 - Understand transformer-based NLP workflows
 - Understand object detection pipeline
 - Understand time-series data characteristics
 - Study real-world system architecture
-

Weekly Self-Check Questions

- Why combine CNN + LSTM in image captioning?
 - Why YOLO is used for real-time detection?
 - Why transformers are best for summarization?
 - Why BERT works well for text classification?
 - When to use LSTM vs ARIMA in time series?
-

Where These Advanced Projects Are Used

- Autonomous Systems
 - Smart Surveillance
 - AI Content Generation
 - Financial Forecasting
 - News & Media Verification Systems
-

Mini Project

Choose one advanced project and:

- Explain complete pipeline (Data → Model → Output)
- Explain why those models are used together
- Explain real-world business or industry usage

Goal: Build advanced AI solution architecture thinking.

End of Week Outcome

You should be able to:

- Understand advanced AI system design
- Understand multi-model deep learning pipelines
- Understand real production-level AI applications
- Prepare for specialization or deployment stage

Week 20 — Specialization Track: Natural Language Processing (NLP)

Weekly Goal

Understand how AI models process, understand, and generate human language using NLP techniques and transformer-based models.

Topics & Subtopics to Complete

1. Text Preprocessing

- Tokenization
 - Stemming
 - Lemmatization
 - Stop Words Removal
-

2. Feature Extraction

- Bag of Words (BoW)
 - TF-IDF
-

3. Word Embeddings

- Word2Vec
 - GloVe
 - FastText
-

4. Sequence Models

- RNN
 - BiRNN
 - LSTM
-

5. Transformers

- BERT
 - GPT
 - Attention Mechanism
-

6. NLP Tasks

- Text Classification
 - Named Entity Recognition (NER)
 - Text Summarization
 - Language Translation
 - Question Answering (QA)
-

Suggested Learning Resources

YouTube

- Krish Naik — NLP Full Playlist
<https://www.youtube.com/c/KrishNaik>
 - StatQuest — NLP Concepts
<https://www.youtube.com/c/joshstarmer>
 - Hugging Face — NLP Course Videos
<https://www.youtube.com/c/HuggingFace>
-

Helpful Websites

- Hugging Face NLP Course
<https://huggingface.co/learn>
 - NLTK Documentation
<https://www.nltk.org>
 - SpaCy Documentation
<https://spacy.io>
 - Kaggle NLP Datasets
<https://www.kaggle.com>
-

Practice Tasks

- Understand text preprocessing pipeline
- Compare BoW vs TF-IDF
- Understand embeddings vs traditional features
- Study transformer-based NLP workflow
- Understand real-world NLP system pipeline

Weekly Self-Check Questions

- Why text preprocessing is important?
 - Difference between TF-IDF and embeddings?
 - Why transformers are best for NLP now?
 - When to use LSTM vs Transformers?
 - How NLP models understand context?
-

Where NLP is Used

- Chatbots & Virtual Assistants
 - Search Engines
 - Translation Systems
 - Sentiment Analysis Systems
 - Document Processing AI
-

Mini Project

Choose one NLP task and:

- Explain full NLP pipeline
- Explain feature extraction method
- Explain model selection

Goal: Build NLP system design understanding.

End of Week Outcome

You should be able to:

- Understand full NLP pipeline
- Understand text feature representation
- Understand transformer-based NLP models
- Prepare for NLP project building or deployment

Week 21 — Specialization Track: Computer Vision (CV)

Weekly Goal

Understand how AI models process images, detect objects, and segment visual data using Computer Vision and Deep Learning architectures.

Topics & Subtopics to Complete

1. Image Basics

- Pixels
 - Color Channels (RGB, Grayscale)
 - Image Formats (JPEG, PNG, etc.)
-

2. Image Processing Techniques

- Thresholding
 - Edge Detection
 - Morphological Operations
-

3. CNN Architectures

- LeNet
 - AlexNet
 - VGG
 - ResNet
 - EfficientNet
-

4. Object Detection

- YOLO
 - SSD
 - Faster-RCNN
-

5. Image Segmentation

- Semantic Segmentation
 - Instance Segmentation
 - U-Net
 - Mask-RCNN
-

Suggested Learning Resources

YouTube

- Krish Naik — Computer Vision Playlist
<https://www.youtube.com/c/KrishNaik>
 - freeCodeCamp — Computer Vision Course
<https://www.youtube.com/c/Freecodecamp>
 - Yannic Kilcher — CV Research Concepts
<https://www.youtube.com/c/YannicKilcher>
-

Helpful Websites

- OpenCV Documentation
<https://opencv.org>
 - PyTorch Vision Tutorials
<https://pytorch.org/tutorials>
 - Papers With Code (CV Models)
<https://paperswithcode.com>
 - Kaggle CV Datasets
<https://www.kaggle.com>
-

Practice Tasks

- Understand image data structure
 - Compare CNN architectures conceptually
 - Understand detection vs segmentation difference
 - Study real-world CV pipeline
 - Understand preprocessing steps for images
-

Weekly Self-Check Questions

- Why CNN is best for image processing?
 - Difference between object detection and segmentation?
 - Why ResNet improved deep CNN training?
 - Why YOLO is used for real-time detection?
 - How segmentation is used in real-world AI?
-



Where Computer Vision is Used

- Face Recognition Systems
 - Autonomous Vehicles
 - Medical Image Analysis
 - Surveillance Systems
 - AR / VR Systems
-



Mini Project

Choose one CV task and:

- Explain image → model → output pipeline
- Explain architecture choice
- Explain real-world usage

Goal: Build Computer Vision system understanding.



End of Week Outcome

You should be able to:

- Understand image-based AI pipelines
- Understand CNN architecture evolution
- Understand detection and segmentation workflows
- Prepare for CV project building

Week 22 — Specialization Track: Generative AI & LLMs



Weekly Goal

Understand how Large Language Models and Generative AI systems generate text, images, and content using transformers, fine-tuning, and prompt engineering.



Topics & Subtopics to Complete

1. LLM Basics

- Tokenization
- Embeddings
- Transformer Blocks

2. GPT Architecture

- Decoder-Only Architecture
 - Causal Attention
 - Autoregressive Generation
-

3. Training LLMs

- Masked Language Modeling (Concept)
 - Next Token Prediction
-

4. Fine-Tuning Techniques

- LoRA
 - QLoRA
 - PEFT
-

5. Prompt Engineering

- Few-Shot Prompting
 - Chain-of-Thought Prompting
 - Instruction Tuning
-

6. Diffusion Models (Image Generation)

- Denoising Process
 - Latent Diffusion
 - Stable Diffusion (Concept Level)
-



Suggested Learning Resources

YouTube

- Hugging Face — LLM & Transformers
<https://www.youtube.com/c/HuggingFace>
 - Andrej Karpathy — LLM & GPT Concepts
<https://www.youtube.com/c/AndrejKarpathy>
 - Yannic Kilcher — LLM Research Concepts
<https://www.youtube.com/c/YannicKilcher>
-

Helpful Websites

- Hugging Face LLM Course
<https://huggingface.co/learn>
 - OpenAI Documentation (Concept Reference)
<https://platform.openai.com/docs>
 - Papers With Code (LLM Research)
<https://paperswithcode.com>
 - Stability AI (Diffusion Concepts)
<https://stability.ai>
-

Practice Tasks

- Understand tokenization workflow
 - Compare embeddings vs traditional features
 - Understand transformer attention flow
 - Study prompt engineering examples
 - Understand diffusion generation concept
-

Weekly Self-Check Questions

- Why transformers are used in LLMs?
 - How GPT generates text step-by-step?
 - Why fine-tuning is needed?
 - Why prompt engineering is powerful?
 - How diffusion models generate images?
-

Where Generative AI & LLMs Are Used

- Chatbots & AI Assistants
 - Content Generation Systems
 - Code Generation AI
 - Image Generation Systems
 - Document Automation AI
-

Mini Project

Choose one Generative AI use-case and:

- Explain full pipeline (Input → Model → Output)
- Explain model selection
- Explain business use-case

Goal: Build Generative AI system understanding.

End of Week Outcome

You should be able to:

- Understand LLM architecture basics
- Understand prompt engineering importance
- Understand generative AI pipelines
- Prepare for GenAI project implementation

Week 24 — Competitive Exposure & Portfolio Building

Weekly Goal

Build strong public proof of skills through competitions, coding platforms, and professional portfolio presence.

Topics & Areas to Complete

Practice & Compete

1. Kaggle

- Participate in Competitions
 - Practice on Real Datasets
 - Target: Competitor → Expert (Long Term Goal)
-

2. AI Hackathons

- Zindi
- Devpost
- Omdena

Focus:

- Real Problem Solving
 - Team Collaboration
 - Real-World AI Exposure
-

3. ML Coding Practice

- LeetCode (ML + Python + Logic Problems)
 - StrataScratch (Data + SQL + ML Problems)
-

**Build Public AI Profile****1. GitHub**

Share:

- ML / DL Project Code
 - Notebooks
 - Model Deployment Projects
 - Documentation (README Files)
-

2. Medium / Hashnode

Share:

- Project Explanations
 - Learning Insights
 - AI Concept Blogs
-

3. LinkedIn

Share:

- Weekly Learning Updates
 - Project Demonstrations
 - AI Learning Journey
-

4. Hugging Face

Share:

- Trained Models
 - NLP / LLM Projects
 - Demo Spaces
-

Suggested Learning Resources

YouTube

- Krish Naik — Portfolio & ML Career Guidance
<https://www.youtube.com/c/KrishNaik>
 - Data Professor — ML Portfolio Projects
<https://www.youtube.com/c/DataProfessor>
 - Tina Huang — AI Career & Portfolio
<https://www.youtube.com/c/TinaHuang1>
-

Helpful Websites

- Kaggle
<https://www.kaggle.com>
 - Devpost (Hackathons)
<https://devpost.com>
 - Zindi (AI Competitions)
<https://zindi.africa>
 - Hugging Face
<https://huggingface.co>
 - GitHub
<https://github.com>
-

Practice Tasks

- Create GitHub profile and upload projects
 - Participate in at least 1 Kaggle competition
 - Write 1 technical blog about project
 - Share 1 project post on LinkedIn
 - Explore 1 Hackathon platform
-

Weekly Self-Check Questions

- Why portfolio is more important than certificates?
- How Kaggle improves real ML skills?

- Why public project sharing helps career growth?
 - Why community participation matters?
 - Why documenting learning is important?
-

Where This Helps in AI Career

- Job Interviews
 - Freelancing Opportunities
 - Startup Opportunities
 - Research Collaboration
 - Personal Branding in AI
-

Mini Project

Build your public AI presence plan:

- List projects to upload on GitHub
- Decide 2 platforms to stay active
- Plan first portfolio blog or post

Goal: Build visible AI career footprint.

End of Week Outcome

You should be able to:

- Build strong AI portfolio
- Compete in real ML challenges
- Present AI skills publicly
- Prepare for job interviews and industry exposure

Week 25 — Job Preparation (AI / ML High-Paying Roles)

Weekly Goal

Prepare for AI/ML job interviews by mastering system design, theory, coding, project discussions, and research-level understanding.

Topics & Areas to Complete

1. ML System Design

- Data Pipelines
 - Feature Stores
 - Model Serving APIs
 - Model Monitoring Systems
-

2. Technical Interviews (AI / ML Theory)

- Machine Learning Algorithms
 - Deep Learning Concepts
 - Mathematics for ML / DL
 - Project-Based Questions
 - Model Selection & Tradeoffs
-

3. Coding Interviews

- Data Structures
- Algorithms
- Problem Solving Patterns
- ML Logic Coding Questions

Practice Platforms:

- LeetCode
 - Codeforces
-

4. Behavioral Interviews

- STAR Method (Situation, Task, Action, Result)
 - Team Collaboration Examples
 - Problem Solving Stories
 - Product Thinking Mindset
-

5. Research Paper Understanding

- Read Core AI Papers
- Understand Model Architecture
- Try Concept-Level Implementation

Examples:

- Attention Is All You Need
 - ResNet Paper
-

Suggested Learning Resources

YouTube

- Krish Naik — ML Interview Preparation
<https://www.youtube.com/c/KrishNaik>
 - AI Coffee Break — Research Papers Explained
<https://www.youtube.com/c/AICoffeeBreak>
 - DataTalksClub — ML System Design
<https://www.youtube.com/c/DataTalksClub>
-

Helpful Websites

- LeetCode
<https://leetcode.com>
 - Papers With Code
<https://paperswithcode.com>
 - arXiv (Research Papers)
<https://arxiv.org>
 - ML System Design Resources
<https://github.com>
-

Practice Tasks

- Solve ML interview questions
 - Practice coding problems daily
 - Prepare project explanations
 - Read at least 1 research paper summary
 - Prepare STAR-based answers
-

Weekly Self-Check Questions

- Can you explain ML project end-to-end?
 - Can you explain model selection logic?
 - Can you solve medium coding problems?
 - Can you explain ML system architecture?
 - Can you discuss tradeoffs in model design?
-



Where This Preparation Helps

- AI Engineer Roles
 - ML Engineer Roles
 - Applied Scientist Roles
 - Data Scientist (Advanced ML)
 - AI Startup Roles
-



Mini Project

Prepare a Mock AI Interview Kit:

- Pick 1 ML project → Explain end-to-end
- Solve 5 coding problems
- Explain 1 ML algorithm deeply
- Summarize 1 research paper

Goal: Build real interview readiness.



End of Week Outcome

You should be able to:

- Explain ML systems confidently
- Solve coding interview questions
- Discuss ML / DL theory clearly
- Demonstrate project expertise in interviews



Bonus — Tools & Libraries You Must Master



Goal

Know the core tools used across the AI lifecycle — from data handling → model building → deployment → production scaling.

Tools & Categories

Language

Python

Machine Learning / Deep Learning

- scikit-learn
 - XGBoost
 - PyTorch
 - TensorFlow
 - Keras
-

Data Processing

- pandas
 - NumPy
-

Visualization

- Matplotlib
 - Seaborn
 - Plotly
-

Computer Vision / NLP

- OpenCV
 - NLTK
 - spaCy
 - Hugging Face Transformers
-

Deployment

- Flask
- FastAPI

- Docker
 - Streamlit
-

Cloud / MLOps

- AWS / GCP
 - MLflow
 - Kubernetes
 - Weights & Biases (W&B)
 - DVC
-

Suggested Learning Resources

YouTube

- Krish Naik — Python, ML, MLOps Tools
<https://www.youtube.com/c/KrishNaik>
 - freeCodeCamp — Full Tool Tutorials
<https://www.youtube.com/c/Freecodecamp>
 - Hugging Face — Transformers & NLP Tools
<https://www.youtube.com/c/HuggingFace>
-

Helpful Websites

- Scikit-Learn Docs → <https://scikit-learn.org>
 - PyTorch Docs → <https://pytorch.org>
 - TensorFlow Docs → <https://tensorflow.org>
 - Hugging Face → <https://huggingface.co>
 - AWS ML → <https://aws.amazon.com/machine-learning>
-

Practice Tasks

- Understand where each tool is used in pipeline
 - Compare PyTorch vs TensorFlow conceptually
 - Understand MLflow / W&B purpose
 - Understand Docker role in deployment
-

Self-Check Questions

- Why Python dominates AI ecosystem?
 - Difference between scikit-learn vs PyTorch?
 - Why deployment tools are important?
 - Why experiment tracking matters?
-

Where These Tools Are Used

- End-to-End AI Development
 - Production AI Systems
 - ML Experimentation
 - Cloud AI Deployment
 - Enterprise AI Applications
-

Mini Project

Create your AI Tool Stack Plan:

- Choose ML framework (PyTorch / TensorFlow)
- Choose deployment stack
- Choose experiment tracking tool

Goal: Understand real AI tech stack selection.

Bonus Outcome

You should be able to:

- Recognize industry-standard AI tools
- Understand where each tool fits in pipeline
- Prepare your personal AI tech stack

Bonus — What to Learn After Math (AI Learning Summary Path)

Goal

Clear high-level roadmap of what comes after Math before becoming a complete AI Engineer.

Learning Flow After Math

Stage 1 — Classical Machine Learning

Learn ML algorithms and apply them on real datasets.

Stage 2 — Deep Learning

Learn Neural Networks → CNN → RNN → Transformers.

Stage 3 — Real AI Projects

Build projects from beginner → intermediate → advanced level.

Stage 4 — Specializations

Choose 1–2 domains:

- NLP
 - Computer Vision
 - LLMs / Generative AI
-

Stage 5 — MLOps & Deployment

Learn how to deploy and scale AI models in production.

Stage 6 — Competitive Exposure & Portfolio

Participate in Kaggle, Hackathons, and build public portfolio.

Stage 7 — Job Preparation

Prepare for system design, coding interviews, ML interviews, and real-world AI roles.

Bonus Outcome

You should be able to:

- Understand complete AI learning journey after math

- Know what to learn at each stage
- Follow structured path without confusion

Week 26 — Data Engineering Foundations (Backbone of Real AI Systems)

Weekly Goal

Understand how real-world AI systems collect, process, store, and deliver data at scale before model training.

Topics & Subtopics to Complete

1. Data Pipelines

- ETL (Extract, Transform, Load)
 - ELT Workflows
 - Data Ingestion Systems
 - Batch vs Scheduled Pipelines
-

2. Data Warehousing

- Snowflake
 - Google BigQuery
 - AWS Redshift
 - Analytical Storage Concepts
-

3. Batch vs Stream Processing

- Batch Processing (Offline Data Processing)
 - Stream Processing (Real-Time Data)
 - Kafka (Event Streaming Concept)
 - Apache Beam (Pipeline Concept)
 - Spark Streaming (Concept Level)
-

4. Querying & Data Access

- SQL Joins
 - Window Functions
 - Common Table Expressions (CTEs)
-

5. Data Versioning

- DVC (Data Version Control)
 - Delta Lake (Concept Level)
-

6. Data Validation & Quality

- Great Expectations
 - Deequ
 - Data Quality Checks
-



Suggested Learning Resources

YouTube

- Data Engineering Zoomcamp (DataTalksClub)
<https://www.youtube.com/c/DataTalksClub>
 - freeCodeCamp — Data Engineering Concepts
<https://www.youtube.com/c/Freecodecamp>
 - TechWorld with Nana — Kafka / Data Systems
<https://www.youtube.com/c/TechWorldwithNana>
-



Helpful Websites

- Snowflake Docs
<https://www.snowflake.com>
- BigQuery Docs
<https://cloud.google.com/bigquery>
- Apache Kafka Docs
<https://kafka.apache.org>

- Delta Lake Docs
<https://delta.io>
-

Practice Tasks

- Understand ETL pipeline flow
 - Study difference between warehouse vs database
 - Understand batch vs real-time data processing
 - Practice SQL advanced queries conceptually
 - Study data quality validation workflow
-

Weekly Self-Check Questions

- Why data pipelines are critical for AI systems?
 - Difference between ETL and ELT?
 - When to use batch vs streaming?
 - Why data versioning is important?
 - Why data quality affects model performance?
-

Where Data Engineering is Used

- Real-Time AI Systems
 - Recommendation Engines
 - Fraud Detection Systems
 - Enterprise AI Platforms
 - Data-Driven AI Products
-

Mini Project

Design a real-world AI data pipeline:

- Data Source → Pipeline → Storage → Model Training
- Explain each stage role

Goal: Build data infrastructure thinking for AI.

End of Week Outcome

You should be able to:

- Understand data flow in real AI systems
- Understand data storage & processing choices
- Understand importance of data quality & versioning
- Prepare for production AI data pipelines

Bonus — AI Ethics, Fairness & Privacy (Enterprise AI Requirement)

Goal

Understand how to build responsible, explainable, and legally compliant AI systems used in real enterprise and global AI products.

Areas & Topics to Know

Bias in AI

- Dataset Imbalance
 - Label Bias / Label Skew
 - Fairness Metrics (Concept Level)
-

Interpretability

- SHAP (Feature Importance Explanation)
 - LIME (Local Model Explanation)
 - Attention Visualization (Concept Level)
-

Privacy in AI

- Differential Privacy (Concept)
 - PII Masking
 - Secure Data Handling Basics
-

Explainable AI (XAI)

- Model Transparency
 - Stakeholder Trust
 - Decision Explanation Importance
-

Legal & Compliance Awareness

- GDPR (Europe Data Protection)
 - HIPAA (Healthcare Data Protection)
 - EU AI Act (Concept Awareness)
 - DPDP (India Data Protection Law)
-

Suggested Learning Resources

YouTube

- StatQuest — Explainable AI Concepts
<https://www.youtube.com/c/joshstarmar>
 - IBM Technology — AI Ethics
<https://www.youtube.com/c/IBMTechology>
 - DeepLearning.ai — Responsible AI
<https://www.deeplearning.ai>
-

Helpful Websites

- Google Responsible AI
<https://ai.google/responsibilities>
 - IBM AI Ethics Resources
<https://www.ibm.com/ai/ethics>
 - EU AI Act Overview
<https://artificialintelligenceact.eu>
 - NIST AI Risk Management
<https://www.nist.gov/itl/ai-risk-management-framework>
-

Practice Tasks

- Study examples of biased datasets
 - Understand why explainability is needed in AI
 - Study real-world AI privacy failures
 - Understand responsible AI development workflow
-

Self-Check Questions

- Why bias is dangerous in AI systems?
 - Why explainability is needed for enterprise AI?
 - Why privacy laws affect AI model design?
 - How AI ethics affects real product deployment?
-

Where This Is Used

- Enterprise AI Systems
 - Healthcare AI
 - Finance AI Systems
 - Government AI Systems
 - Large Scale Consumer AI Products
-

Mini Project

Choose one AI system (like loan approval or hiring AI) and:

- Identify possible bias risks
- Explain how to make it explainable
- Explain privacy protection approach

Goal: Build responsible AI thinking.

Bonus Outcome

You should be able to:

- Understand responsible AI development
- Understand AI fairness & bias risks
- Understand privacy and compliance basics
- Prepare for enterprise-level AI roles

Bonus — Paper Reading & AI Research Skills (Research-Driven AI Roles)

Goal

Develop the ability to read, understand, and implement state-of-the-art AI research papers used in advanced AI roles and research labs.

Areas & What to Do

Reading ML / DL Research Papers

- Learn how to read arXiv papers
 - Focus on: Problem → Method → Results → Limitations
 - Start with Transformers, Diffusion Models, Vision Models
-

Reproducing Research Results

- Try concept-level implementation
 - Study official GitHub repos
 - Implement simplified versions using PyTorch / TensorFlow
-

Tools for Research Learning

- Papers With Code (Paper + Code + Benchmarks)
 - arXiv Sanity (Paper Discovery & Filtering)
-

Core Papers to Start With

Transformers

- Attention Is All You Need

Language Models

- BERT — Pre-training of Deep Bidirectional Transformers

Generative AI

- DALL·E
- Stable Diffusion

Scientific AI

- AlphaFold
- AlphaGo

Computer Vision Foundation Models

- Segment Anything (Meta AI)
-

Suggested Learning Resources

YouTube

- Yannic Kilcher — Research Paper Explanations
<https://www.youtube.com/c/YannicKilcher>
 - AI Coffee Break — Paper Simplifications
<https://www.youtube.com/c/AICoffeeBreak>
 - Two Minute Papers — Research Summaries
<https://www.youtube.com/c/KárolyZsolnai>
-

Helpful Websites

- arXiv (Research Papers)
<https://arxiv.org>
 - Papers With Code
<https://paperswithcode.com>
 - arXiv Sanity
<http://www.arxiv-sanity.com>
-

Practice Tasks

- Read 1 research paper summary per week
 - Watch paper explanation videos
 - Try understanding model architecture diagrams
 - Study benchmark comparisons
-

Self-Check Questions

- Can you explain a research paper in simple terms?
 - Can you explain why the method works?
 - Can you explain advantages vs older methods?
 - Can you identify real-world applications of research?
-

Where These Skills Are Used

- AI Research Labs
 - Advanced AI Product Teams
 - Applied Research Roles
 - PhD / Research-Oriented Careers
 - Frontier AI Startups
-

Mini Project

Pick one research paper and:

- Explain problem solved
- Explain model architecture
- Explain why it is important

Goal: Build research thinking mindset.

Bonus Outcome

You should be able to:

- Understand how to read ML / DL research papers
- Navigate AI research ecosystem
- Understand new AI breakthroughs faster
- Prepare for research or advanced AI roles

Bonus — Advanced Mathematics (Optional but Powerful for AI Research)

Goal

Go beyond basic ML math and understand advanced mathematical foundations used in cutting-edge AI research and architecture design.

Fields & Topics to Learn

Numerical Optimization

- Convexity
- Gradient-Based Methods
- Newton's Method

Used In:

Advanced Model Training • Optimization Theory • Research-Level ML

Information Theory (Advanced Level)

- KL Divergence (Advanced Understanding)
- Mutual Information
- Entropy Bounds

Used In:

LLMs • Generative Models • Compression-Based Learning

Functional Analysis

- Function Spaces
- Convergence of Function Sequences

Used In:

Neural Network Theory • Infinite-Dimensional Learning Models

Topology & Manifold Learning

- Manifold Hypothesis (Concept)
- Representation Learning Geometry

Used In:

Latent Space Modeling • Embeddings • Advanced Representation Learning

Riemannian Geometry

- Optimization on Curved Spaces
- Geometry-Aware Learning

Used In:

Advanced Deep Learning Optimization • Large Model Training Research

Suggested Learning Resources

YouTube

- MIT OpenCourseWare — Advanced Mathematics
<https://www.youtube.com/c/MITOpenCourseWare>
 - 3Blue1Brown — Advanced Math Intuition
<https://www.youtube.com/c/3blue1brown>
 - Yannic Kilcher — AI Research Math Concepts
<https://www.youtube.com/c/YannicKilcher>
-

Helpful Websites

- MIT OCW Advanced Math
<https://ocw.mit.edu>
 - Distill.pub (Math + ML Visualization)
<https://distill.pub>
 - Papers With Code
<https://paperswithcode.com>
-

Practice Tasks

- Study advanced optimization intuition
 - Understand entropy bounds conceptually
 - Study manifold learning visual examples
 - Study geometric interpretation of embeddings
-

Self-Check Questions

- Why convex optimization is important in ML?
 - Why information theory is core to generative AI?
 - Why geometry is becoming important in deep learning?
 - Why advanced math helps in research roles?
-

Where This Is Used

- Advanced AI Research Labs
 - Foundation Model Research
 - AI Architecture Design Roles
 - Frontier AI Startups
 - PhD-Level AI Research
-

Mini Project

Choose one advanced math topic and:

- Explain concept in simple terms
- Explain where it is used in AI
- Explain why it is important for future AI

Goal: Build research-level mathematical thinking.

Bonus Outcome

You should be able to:

- Understand advanced AI math terminology
- Recognize math used in research papers
- Understand theoretical AI foundations
- Prepare for research-oriented AI roles

Week 27 — Prompt Engineering & LLM Ops (Industry Critical Skill)

Weekly Goal

Understand how to design effective prompts, fine-tune LLMs, deploy them in production, and evaluate real-world LLM systems.

Topics & Subtopics to Complete

1. Prompt Engineering

- Zero-Shot Prompting
 - Few-Shot Prompting
 - Chain-of-Thought (CoT)
 - ReAct (Reason + Act)
 - Retrieval-Augmented Generation (Prompt + Knowledge Retrieval Concept)
-

2. LLM Fine-Tuning

- LoRA
 - QLoRA
 - PEFT (Parameter Efficient Fine Tuning)
 - Adapter Layers
-

3. Serving LLMs

- vLLM (Concept Level)
 - Triton Inference Server (Concept Level)
 - High-Throughput Inference
-

4. Retrieval-Augmented Generation (RAG)

- Vector Databases Concept
 - FAISS
 - ChromaDB
 - LangChain RAG Pipelines
-

5. LLM Agents

- LangGraph
 - CrewAI
 - AutoGen
 - Multi-Agent Workflows
-

6. LLM APIs & Ecosystem

- OpenAI APIs
 - Claude APIs
 - Cohere APIs
 - Gemini APIs
 - Mistral APIs
-

7. LLM Evaluation

- Perplexity
 - BLEU Score
 - BERTScore
 - Human Evaluation
-



Suggested Learning Resources

YouTube

- Hugging Face — LLM + Transformers
<https://www.youtube.com/c/HuggingFace>
 - Andrej Karpathy — LLM Deep Dive
<https://www.youtube.com/c/AndrejKarpathy>
 - LangChain — RAG + Agents Tutorials
<https://www.youtube.com/c/LangChain>
-



Helpful Websites

- Hugging Face Learn
<https://huggingface.co/learn>
 - LangChain Docs
<https://docs.langchain.com>
 - FAISS Docs
<https://faiss.ai>
 - OpenAI Platform Docs
<https://platform.openai.com/docs>
-

Practice Tasks

- Compare prompt types (Zero vs Few vs CoT)
 - Study RAG pipeline flow
 - Understand agent-based workflows
 - Understand LLM deployment architecture
 - Study evaluation metric differences
-

Weekly Self-Check Questions

- Why prompt engineering is powerful in GenAI?
 - Why RAG is needed instead of pure LLM prompting?
 - Why parameter-efficient fine-tuning is important?
 - Why LLM evaluation is harder than ML evaluation?
 - Why agents are becoming important in AI apps?
-

Where These Skills Are Used

- AI Chatbots & Assistants
 - Enterprise LLM Applications
 - AI Automation Systems
 - Knowledge Retrieval AI Systems
 - GenAI Product Development
-

Mini Project

Design an LLM Application Pipeline:

- User Query → Prompt → Retrieval (Optional RAG) → LLM → Response
- Explain tool stack choice

Goal: Build real-world LLM system design thinking.

End of Week Outcome

You should be able to:

- Design effective prompts
- Understand LLM fine-tuning workflow

- Understand RAG and agent architectures
- Understand LLM production deployment concepts

Week 28 — Systems Thinking & AI Infrastructure (High-End AI Roles)

Weekly Goal

Understand how large-scale AI systems are designed, optimized, deployed, and scaled using distributed infrastructure and specialized hardware.

Topics & Subtopics to Complete

1. ML System Design (Production Architecture)

- Model Serving Architecture
 - Model Versioning
 - Canary Deployment (Safe Model Rollout Concept)
-

2. Vector Databases

- Vector Search Concept
 - FAISS
 - Pinecone
 - Weaviate
-

3. Distributed Training

- Data Parallelism
 - Model Parallelism
 - Distributed Training Frameworks (Horovod — Concept Level)
-

4. GPU Optimization

- CUDA (Concept Level)
 - cuDNN (Deep Learning Acceleration Concept)
 - Mixed Precision Training
-

5. Hardware Awareness

- GPUs (A100, H100 — Concept Awareness)
 - TPUs (Concept Level)
 - Inference Latency Optimization
 - Batch Size Tuning
-

Suggested Learning Resources

YouTube

- Full Stack Deep Learning — ML Systems
<https://www.youtube.com/c/FullStackDeepLearning>
 - NVIDIA Developer — GPU & CUDA Concepts
<https://www.youtube.com/c/NVIDIADeveloper>
 - TechWorld with Nana — System Architecture Concepts
<https://www.youtube.com/c/TechWorldwithNana>
-

Helpful Websites

- NVIDIA CUDA Docs
<https://developer.nvidia.com/cuda-zone>
 - Pinecone Vector DB Docs
<https://www.pinecone.io>
 - Weaviate Docs
<https://weaviate.io>
 - FAISS Documentation
<https://faiss.ai>
-

Practice Tasks

- Study large-scale ML system architecture diagrams
 - Understand distributed training workflow
 - Understand vector search pipeline
 - Study GPU acceleration benefits
 - Study latency vs throughput tradeoffs
-

Weekly Self-Check Questions

- Why system design is critical for production AI?
 - Why vector databases are needed for LLM systems?
 - When do we need distributed training?
 - Why GPU optimization is critical for large models?
 - How hardware affects model performance?
-

Where This Is Used

- LLM Infrastructure Teams
 - AI Platform Engineering
 - Large-Scale ML Training Systems
 - Enterprise AI Product Architecture
 - High-End AI Research Engineering Roles
-

Mini Project

Design a scalable AI system architecture:

- Data → Training → Model Serving → Vector DB → Monitoring → Users
- Explain scaling strategy

Goal: Build production-scale AI system design thinking.

End of Week Outcome

You should be able to:

- Understand large-scale AI system architecture
- Understand distributed training concepts
- Understand vector database role in GenAI
- Understand hardware-aware AI system design

Week 29 — Soft Skills + Domain Knowledge (Senior AI Engineer / High-Paying Roles)

Weekly Goal

Develop communication, domain understanding, leadership, and documentation skills required for senior AI roles and decision-making positions.

Skills & What to Learn

Communication Skills

- Explaining ML / AI models to non-technical stakeholders
- Business impact storytelling
- Data-driven decision communication

Why It Matters:

AI Engineers must explain results to product, business, and leadership teams.

Domain Expertise

- Finance AI
- Healthcare AI
- Robotics / Automation
- Gaming AI
- E-Commerce AI

Why It Matters:

Domain knowledge helps build practical, production-ready AI solutions.

Project Planning & Execution

- Agile Methodology
- Jira Workflow
- SCRUM Process
- Sprint Planning
- AI Project Lifecycle Planning

Why It Matters:

Real AI projects are team-driven and sprint-based.

Leadership Skills

- Leading ML Teams
- Mentoring Junior Engineers
- Owning AI Product Features
- Technical Decision Making

Why It Matters:

Senior AI roles require ownership, not just coding.

Technical Writing & Documentation

- Model Cards
- Technical Documentation
- README Writing
- Ethical Declarations
- Research Documentation

Why It Matters:

Documentation = Trust + Reproducibility + Compliance.

Suggested Learning Resources

YouTube

- Tina Huang — AI Career Growth & Communication
<https://www.youtube.com/c/TinaHuang1>
 - TechLead / Engineering Leadership Concepts
<https://www.youtube.com>
 - Google AI — Responsible AI Communication
<https://www.youtube.com>
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Helpful Websites

- Atlassian Agile & Scrum Guides
<https://www.atlassian.com/agile>
- Google Model Card Documentation
<https://modelcards.withgoogle.com>

- Microsoft Responsible AI Docs
<https://www.microsoft.com/ai>
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Practice Tasks

- Explain one ML project in simple business terms
 - Pick one industry and study AI use cases
 - Write one technical README for project
 - Study Agile workflow for AI projects
 - Practice writing model explanation summary
-

Weekly Self-Check Questions

- Can you explain your AI model to business team?
 - Do you understand one industry deeply?
 - Can you plan an AI project roadmap?
 - Can you document model decisions clearly?
 - Can you justify AI design choices?
-

Where These Skills Are Used

- Senior AI Engineer Roles
 - AI Tech Lead Roles
 - AI Product Roles
 - AI Research Leadership Roles
 - AI Startup Founder Roles
-

Mini Project

Create your AI Professional Profile Plan:

- Pick 1 domain specialization
- Write 1 model explanation for non-tech audience
- Write 1 project documentation sample

Goal: Build AI leadership + real-world communication thinking.

✓ End of Week Outcome

You should be able to:

- Communicate AI concepts clearly
- Connect AI with real industry problems
- Document AI systems professionally
- Think like a senior AI engineer / decision maker

Week 30 ✓ Final Checklist: After Math, You Should Learn...

Stage	Area
1	Classical Machine Learning
2	Deep Learning
3	Project Building
4	NLP, CV, LLM, or GenAI specialization
5	MLOps (deployment, monitoring, scaling)
6	Competitive profile (Kaggle, GitHub, blogs)
7	Interview + System Design
8	Data Engineering
9	Ethics, Bias, Privacy, XAI
10	Paper Reading & Research
11	Advanced Mathematics (optional but great)
12	Prompt Engineering & LLMOps
13	ML Systems / Infra Design
14	Soft skills + Domain knowledge

1 Evolve with the Field (Stay Updated Weekly)

AI is changing **monthly**, so *staying current is now a skill*.

↻ How to do it:

- Read **arXiv** or **PapersWithCode** weekly
- Track **Hugging Face**, **OpenAI**, **Meta AI**, **Google DeepMind**
- Follow newsletters:
 - *The Batch* (by Andrew Ng)
 - *Import AI* (Jack Clark)
 - *Latent Space* (for LLMs)
- Explore trending repos on GitHub

💡 Knowledge **decays fast** in AI — adapt fast, stay relevant.

2 Get Real-World Experience (The Only Real Teacher)

Learn by doing:

- Join or contribute to **open-source AI projects**
- Work in teams, handle messy data, design infra
- Build full-stack AI products (UI + backend + LLMs)
- Intern or consult for startups (part-time if needed)
- Work on real problems in healthcare, finance, e-commerce, etc.

🎯 **10 real-world projects > 10,000 pages of theory.**

3 Go Deep in One Subdomain (Become a Specialist)

You can't master **everything** at 40+ LPA level — but you **can** be world-class in one of:

Domain	Why it Matters
✅ NLP / LLMs	Dominates 2025 AI job market
✅ Vision / Robotics	Hardware + AI = big market

✓ Generative AI	Models for images, video, music
✓ BioAI / MedAI	Personalized medicine, protein folding
✓ Autonomous Systems	Self-driving, drones
✓ FinTech AI	Risk modeling, algo trading
✓ Multi-modal AI	Text + image + audio models
✓ Agentic AI	Autonomous LLM agents & planning systems

Pick one, **go deep**, and **solve real problems** in it.

Final Truth: After a Point, Learning ≠ Reading

Once you've completed the roadmap, the next learning comes from:

- **Solving bigger, real-world problems**
 - **Leading teams or systems**
 - **Contributing to open research**
 - **Building your own AI products**
 - **Teaching, mentoring, writing**
-

You're Now in the "Execution Phase"

Everything beyond this point is about:

-  **Creating real value**
-  **Making an impact**
-  **Innovating**
-  **Earning from your skill**
-  **Mastering your niche**

Join in Your Telegram Link- <https://t.me/+RAC2BZuu9Dg1ZGI1>
You are not Alone here.

mahi community